

Fidel González Leiva

Robotics and Automation Engineer — Software Developer

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PROFILE

Software developer with 6+ years of experience building C++ and Python systems for automotive, robotics, and computer-vision applications. Experienced in AUTOSAR, ROS, real-time processing, build-system modernization, automated testing, and cross-functional delivery in safety-critical engineering environments.

EDUCATION

University of Seville <i>M.Sc. in Electronic, Robotics and Automation Engineering</i>	Seville, Spain 2018 – 2019
UMA <i>B.Sc. in Electronic Robotic and Mechatronic Engineering</i>	Málaga, Spain 2014 – 2017

SKILLS

Languages: C++11/14/17, C, Python, Bash, SQL, Go, JavaScript/TypeScript

Build & Tools: GCC, Clang/LLVM, Git, Make, Bazel, Docker, GDB, Jenkins, ClearCase, Wireshark

Systems & Frameworks: STL, AUTOSAR, multithreading, IPC, GTest/GMock, ROS, OpenCV, SQLite, PostgreSQL

Practices: Agile, Scrum, CI/CD, Requirement-Driven Development, TDD

EXPERIENCE

Alten Delivery Center Spain — ASML <i>Software Developer</i>	Remote, Spain Feb 2025 – Present
- Led technical-debt assessments across a large C/C++ codebase and production libraries, identifying circular dependencies, duplicated symbols, and inconsistent dependency declarations. - Modernized build infrastructure by migrating large lithography-machine components from Makefiles to Bazel, improving dependency management and build reproducibility. - Served as AI Champion, evaluating relevant tools and establishing practical workflows that enabled developers to adopt AI-assisted engineering practices. - Built automation scripts and verification tooling that reduced manual intervention and improved consistency across development and maintenance activities. - Improved code-review and engineering practices by contributing to more efficient development workflows and quality processes.	
Career Break <i>Personal & Family Responsibilities</i>	Spain May 2024 – Feb 2025
Family caregiving and personal responsibilities.	
Bertrandt Group — BMW <i>Software Developer</i>	Seville, Spain Jul 2022 – May 2024
- Designed software architecture for extendable door handles in next-generation BMW vehicles, maintaining traceability between system states, events, and requirements. - Developed automotive-critical C++ software with AUTOSAR and applied Requirement-Driven Development throughout the software lifecycle. - Diagnosed integration issues by analyzing CAN, LIN, SOME/IP, and DLT traces across interdependent components. - Managed Git-based integration and developed automated unit tests with GTest/GMock for automotive software.	
FADA-CATEC <i>Robotics Engineer — Software Developer</i>	Seville, Spain Jun 2019 – Jul 2022
- Developed real-time C++/Python software for autonomous UAVs, covering computer vision, AI-based perception, navigation, and mission execution. - Designed computer vision algorithms for aerial target detection, tracking, and interception, contributing to research published at ICUAS 2020. - Developed an AI-based aircraft detection system for BVLOS operations, resulting in a peer-reviewed publication in <i>Electronics</i>. - Contributed to multi-UAV systems for urban operations, 5G-enabled security, and disaster-response applications.	

PROJECTS

AI Agentic Programming | *Agent loops, memory, and tool use*

Mar 2026 – May 2026

- Built and documented agent-loop experiments covering memory, tool use, orchestration, and reasoning workflows.
- Created technical presentations introducing developers to agentic AI concepts and practical implementation patterns.
- Repository: <https://github.com/lotape6/FindingMemo>

Cybersecurity | *Red Team / Blue Team*

Sep 2024 – Jan 2025

- Practiced penetration testing, privilege escalation, web exploitation, and Active Directory enumeration across Windows and Linux labs.
- Completed blue-team labs and LetsDefend investigations covering alert triage, phishing analysis, IOC validation, malware analysis, endpoint investigation, and incident response.
- Repository: <https://github.com/lotape6/HackingNotebook>

Artificial Intelligence | *NLP / LLM Models*

Sep 2024 – Nov 2024

- Studied LLMs and built a local RAG-based agent using Python, PGVector, and LlamaIndex.
- Repository: <https://github.com/lotape6/AIgen>

PUBLICATIONS

Autonomous drone with ability to track and capture an aerial target

2020

- M. García, R. Caballero, F. González, A. Viguria, and A. Ollero. In *2020 International Conference on Unmanned Aircraft Systems (ICUAS)*, pp. 32–40.

Audio-Based Aircraft Detection System for Safe RPAS BVLOS Operations

2020

- J. Mariscal-Harana, V. Alarcón, F. González, J. J. Calvente, F. J. Pérez-Grau, A. Viguria, and A. Ollero. *Electronics*, 9(12), 2076.

Vision-based UAV Detection for Air-to-Air Neutralization

2021

- F. González, R. Caballero, F. J. Pérez-Grau, and A. Viguria. In *2021 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)*, pp. 236–241.